ICT378 TSA 2025
Cyber Forensics & Incident Response

Assignment Information

You **must submit your assignment online using the Assignment course tool**.

You must submit your assignment as **ONE** word-processed document containing **all** of the required question answers.

You **must** keep a copy of the final version of your assignment as submitted and be prepared to provide it on request.

The University treats plagiarism, collusion, theft of other students' work and other forms of dishonesty in assessment seriously. For guidelines on honesty in assessment including avoiding plagiarism, see:
http://www.murdoch.edu.au/Curriculum-and-Academic-Policy/Student-Integrity/

2019 Narcos

Due to intelligence provided by the Australian government, two passengers were intercepted by Customs upon arriving in Wellington, New Zealand from Brisbane. The Intel stated that Jane Esteban and John Fredricksen may be involved in illegal activity.

The suspects were each searched by a customs officer. John Fredricksen's baggage consisted of clothing, toiletries and a Windows laptop. Jane Esteban's baggage also consisted of clothing, toiletries and a small windows laptop.

Upon further search of the lining of the suitcase, one kilogram of Methamphetamine was located. Both suspects were taken into separate interview rooms where they were interrogated. John Fredricksen refused to answer any questions.

Jane Esteban stated all she knew was that she had to deliver the suitcase to the Eastbourne library but if all else failed then they were to deliver it to 666 Rewera Avenue, Petone as told by John.

Customs and police subsequently raided that address. There was nobody present at the address. Customs did, however, find drugs, guns and a desktop computer in the living room of the suspect's house.

You are a Customs forensics investigator. Customs officers have delivered images and memory dumps of the 2 laptops and 1 desktop computer to you. Your task is to carry out a forensic examination of John Fredricksen, Jane Esteban and the unknown suspect's laptops and desktop computers to further understand their motives, goals and objectives. It should be noted that all three devices contain different Windows 10 builds and resulting artefacts may not be located in the same location or even be present.

Suspects Descriptions

Personas:

John Fredricksen

John has been communicating with Steve Kowhai (NZ dealer) via with what he believes is a secure and private chat room (Discord) to discuss his new consignment. Their chat contains information on where they are going and what he wants John Fredricksen to deliver. Furthermore, Steve shares some documents via (email, cloud, etc) that will assist with his job.

John Fredricksen now has enough information to concoct his plan of smuggling the 1kg of methamphetamine into New Zealand, but he needs to find some cover that can take the heat off of himself if any surprises were to happen. John identifies Jane Esteban as a regular user of his business's product (meth) and thinks she will make a great mule for smuggling the drugs.

Jane Esteban

Jane is an undercover Australian Federal Police (AFP) officer tasked with gathering evidence about a drug ring involving John Fredricksen and his associate Steve Kowhai in New Zealand.

Jane will be using the following persona while working undercover. She has a terrible addiction and has been visiting John to feed her addiction, which has led to a transactional friendship with him as a result. John approaches Jane soon after discussing with Steve to convince her to assist with his job.

Steve Kowhai

Steve is a big player drug distributor/dealer in the lower north island of New Zealand and is wanting to find some quality product to expand his growing empire even more. Steve has contacted a source (John) in the US to smuggle in a taster of the product he plans to buy in larger quantities later. Steve has provided John with information about New Zealand and points on how best to smuggle the product into Wellington without raising any alarms at customs. Steve knows a thing or two about digital forensics and decided to use steganography to hide the document within a picture.

Materials – Drive Image

The evidence for this scenario includes Steve Kowhai drive image, Jane Esteban drive image, John Fredricksen drive image, Steve Kowhai memory image, Jane Esteban memory image, and John Fredricksen memory image. The materials can be downloaded from https://downloads.digitalcorpora.org/corpora/scenarios/2019-narcos/

Drive Images		
Actor	**File Name**	**MD5 Hash**
Steve Kowhai	Narcos-1a.001-021	996182c381ec9e7025f40519107615e4
Jane Estaban	Narcos-3a.001-021	ce707bf783dde13ed42196cd6e473083
John Fredricksen	Narcos-2a.001-021	56823dee9b24a40407bec184f80261c2
Memory images		
Actor	**File Name**	**MD5 Hash**
Steve Kowhai	Narcos-Mem-1a.001-003	3469089f9a26b0b51a4ee985cd1c3008
Jane Estaban	Narcos-Mem-3a.001-003	7c155dae658fb059586b3ab5144e21d2
John Fredricksen	Narcos-Mem-2a.001-003	fd7027c7bffedd653226355467c2b1ef

Deliverable Report

Task Description

You should follow forensics procedures, such as taking a hash of the image before using it and checking regularly to ensure you have not modified it. You can select and use any proprietary or open source tools that you have been introduced to or find yourselves to perform the analysis and extract any evidence present.

Your report should detail the investigation process and the findings (including copies of relevant evidence), including obstacles and problems that you encountered and how you overcame them. You can assume that the reader has a light understanding of digital forensics, so any complicated terms/techniques/etc should be explained.

You must include some screenshots in your reports with the output of the tools or the processes and when necessary to support/show how you reached your conclusions. Screenshots should not be used to excess – they merely serve to demonstrate your understanding of the tools/processes and should be used to support written explanations (not in place of).

You will be marked based on the evidence you extract, the use of appropriate tools, the detail of the process, the explanation on its relevance to the case and documentation. Remember, you report should present the information in an unbiased way. Improper handling/validation of evidence would result in loss of marks except where accurately identified and corrected.

**This assignment can be accomplished either individually or in pairs.

Marking Rubric:

The following table summarizes the marking criteria of the final report.

Sections	Marks
Cover Page, Table of Contents, Executive summary	5
Methodology	10
Findings (use of appropriate tools and details of the process) • Discussions (the explanation on findings' relevance to the case) • Supporting Evidence (accurate data acquisition)	65
Summary & Appendix	10
References & Formatting	10
Total	100

Your report must highlight the following requirements (these will be assessed):

A. "Provide a written summary not to exceed two pages that describes what took place":

- Clear and concise summary.
- The summary is objective, not subjective.
- Report only includes relevant artefacts pertaining to the case.
- "The written summary is free of grammatical, syntax, and spelling errors, e.g., consistent verb tense, pronoun-antecedent agreement, correct use of parallelism, etc".

B. "Provide a written description not to exceed four pages of the forensic methodology used to analyse the evidence files and obtain the results identified in the summary. The methodology does not need to provide step-by-step instructions on how software was used; however, it should provide a sufficient description for the findings to be reproduced":

- A methodology explaining the forensic process of how artefacts were identified.
- The methodology is forensically sound and is defensible.
- The methodology is reproducible for other forensic examiners.
- The methodology and table of findings support the conclusions presented in the Narcos scenario.
- "The written methodology is free of grammatical, syntax, and spelling errors, e.g., consistent verb tense, pronoun-antecedent agreement, correct use of parallelism, etc".

C. "Provide a Table of Findings, which contains a list of recovered artefacts with forensic information to build a defensible case". "User-produced evidence files (include name of file(s), MD5 hash, and locations(s) and comment regarding evidentiary value)":

- Identify relevant user account profiles and computer names associated with the suspect's computers.
- Identify relevant web activity on each suspect's computer.
- Identify images that help to build a profile of the three suspects' behaviour.
- Identify binary files that could help the investigation.
- Identify the means and the content of communications between all the suspects.
- Identify any documents that could help the investigation.
- Identify any obfuscation methods used by the suspects.
- Identify encryption methods used by the suspects and determine two methods that can circumvent the encryption.
- Identify malware used by one of the suspects and determine its purpose.
- Identify the vulnerability that allowed the malware to function.
- Leverage other Windows artefacts that provide corroborative evidence such as Windows Timeliner.
- Identify whether changes have occurred to these artefacts across the different Win 10 builds.
- Identify the roles of each suspect.

Sample Structure for Report

Outline: Use the following as a starting point to structure your report

Cover Page

- Title
- Date
- Student Name / Student Number

Table of Contents

- Main contents listed with page number
- Be sure to include visible page numbers on all pages

Executive summary

- Brief Description of the event
- Brief methodology of the investigation
- Brief evidence collection and preservation methods
- Conclusion with short, generalized reasons (like bullet-points)

Methodology details

- Investigation
- Evidence collection and preservation

Finding 1 - Description

- Discussion (e.g. Inculpatory or Exculpatory)
- Supporting evidence

Finding *n* - Description

- Discussion (e.g. Inculpatory or Exculpatory)
- Supporting evidence

Summary and Conclusion

- Discuss if there is there any evidence of illegal drug activity (Methamphetamine).
- How sound / reliable do you believe your evidence collection tobe?
- Is the person innocent or guilty? Explain your position.

Appendix

- Description of persons of interest (often shown in table format)

- Association Diagram of persons of interest
- Evidence listing
- Evidence Timeline (present any evidence in a timeline format, signposting the points where you believe any offence may have occurred and other significant dates/times in the case).
- Software and tools used in the investigation
- Other important listings and information as needed

References:

Your report should be your own, and you should use appropriate citation and referencing formats. All sources that you use as supporting material to your reports must be referenced according to the convention. Failure to do so will result in the loss of marks! You should use APA as a referencing style. The IEEE format is also acceptable.

Formatting:

1. Paragraph text: Font size 12 with Calibri or Times New Roman font. 1.5line spacing. *Justify* alignment (ctrl+j in word).
2. Use Word (or equivalent) styles for headings, paragraphs, etc., to ensureconsistency.
3. Number chapters (1, 2, etc.) and sub-chapters (e.g. 1.1, 2.1, 2.2) – and consistently.
4. Figures should have a figure number and a caption (right click and insert a caption in Word).
5. Write in the third person.
6. Word limit: **maximum 3500 words. Note that the word limit for group work is maximum 5000 words**